BIVARIATE TRUNCATED MOMENT SEQUENCES WITH THE COLUMN RELATION $XY = X^m + q(X)$, WITH q OF DEGREE $m-1$

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Abstract. When the algebraic variety associated with a truncated moment sequence is finite, solving the moment problem follows a well-defined procedure. However, moment problems involving infinite algebraic varieties are more complex and less well-understood. Recent studies suggest that certain bivariate moment sequences can be transformed into equivalent univariate sequences, offering a valuable approach for solving these problems.

In this paper, we address the truncated moment problem for certain rational plane curves. For curves of general degree, we derive an equivalent Hankel positive semidefinite completion problem. For cubic curves, we solve this problem explicitly, thereby resolving the truncated moment problem for one of the four types of irreducible cubic curves, up to affine linear equivalence. For the quartic case, we reduce the completion problem to a feasibility question involving a system of inequalities in three variables. In the cubic case, we also determine the Carathéodory number of the associated moment sequences, complementing recent results on real plane cubics and affine plane curves.

1. Introduction

Given a real 2-dimensional multisequence of degree m, $\beta \equiv \beta^{(m)} = \{\beta_{00}, \beta_{10}, \beta_{01}, \cdots, \beta_{m,0}, \beta_{m-1,1}, \cdots, \beta_{1,m-1}, \beta_{0,m}\}$ with $\beta_{00} > 0$, the **truncated moment problem** (TMP) entails finding necessary and sufficient conditions for the existence of a positive Borel measure μ such that $\operatorname{supp} \mu \subseteq \mathbb{R}^2$ and

$$\beta_{i,j} \equiv \beta_{(i,j)} = \int x^i y^j \, d\mu \;\; (0 \leq i+j \leq m; \;\; i,j \in \mathbb{Z}_+).$$

In this context, we refer to μ as a **representing measure** for β or the moment matrix $M(n)$ defined below. When the order of a moment sequence is even, such as $m = 2n$ for some $n \in \mathbb{N}$, it is possible to define the **moment matrix** $M(n) \equiv M(n)(\beta^{(2n)})$ of β as follows:

$$M(n) \equiv M(n)(\beta^{(2n)}) := (\beta_{\mathbf{i}+\mathbf{j}})_{\mathbf{i},\,\mathbf{j} \in \mathbb{Z}^2_+ : |\mathbf{i}|,\,|\mathbf{j}| \leq n},$$

where $\mathbf{i}+\mathbf{j}$ stands for the coordinate-wise sum and $|\mathbf{i}|$ is the sum of coordinates of $\mathbf{i}$. To guarantee the existence of a representing measure for $\beta^{(2n)}$, it is essential that the matrix $M(n)$ is positive semidefinite. However, there are additional conditions that must be met. To examine these conditions, let $\mathbb{R}[x,y]_n$ denote the set of bivariate polynomials in $\mathbb{R}[x,y]$ with degree at most n. We arrange the columns of

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$M(n)$ according to the monomials $\{1, X, Y, X^2, XY, Y^2, \ldots, X^n, \ldots, Y^n\}$ in degree-lexicographic order. For instance, a sextic moment matrix $M(3)$ would appear as follows:

$$\begin{array}{c} \\ 1 \\ X \\ Y \\ X^2 \\ XY \\ Y^2 \\ X^3 \\ X^2Y \\ XY^2 \\ Y^3 \end{array} \begin{array}{c} \begin{array}{cccccccccc} 1 & X & Y & X^2 & XY & Y^2 & X^3 & X^2Y & XY^2 & Y^3 \end{array} \\ \left(\begin{array}{c|c|c|c|c|c|c|c|c|c} \beta_{00} & \beta_{10} & \beta_{01} & \beta_{20} & \beta_{11} & \beta_{02} & \beta_{30} & \beta_{21} & \beta_{12} & \beta_{03} \\ \hline \beta_{10} & \beta_{20} & \beta_{11} & \beta_{30} & \beta_{21} & \beta_{12} & \beta_{40} & \beta_{31} & \beta_{22} & \beta_{13} \\ \beta_{01} & \beta_{11} & \beta_{02} & \beta_{21} & \beta_{12} & \beta_{03} & \beta_{31} & \beta_{22} & \beta_{13} & \beta_{04} \\ \hline \beta_{20} & \beta_{30} & \beta_{21} & \beta_{40} & \beta_{31} & \beta_{22} & \beta_{50} & \beta_{41} & \beta_{32} & \beta_{23} \\ \beta_{11} & \beta_{21} & \beta_{12} & \beta_{31} & \beta_{22} & \beta_{13} & \beta_{41} & \beta_{32} & \beta_{23} & \beta_{14} \\ \beta_{02} & \beta_{12} & \beta_{03} & \beta_{22} & \beta_{13} & \beta_{04} & \beta_{32} & \beta_{23} & \beta_{14} & \beta_{05} \\ \hline \beta_{30} & \beta_{40} & \beta_{31} & \beta_{50} & \beta_{41} & \beta_{32} & \beta_{60} & \beta_{51} & \beta_{42} & \beta_{33} \\ \beta_{21} & \beta_{31} & \beta_{22} & \beta_{41} & \beta_{32} & \beta_{23} & \beta_{51} & \beta_{42} & \beta_{33} & \beta_{24} \\ \beta_{12} & \beta_{22} & \beta_{13} & \beta_{32} & \beta_{23} & \beta_{14} & \beta_{42} & \beta_{33} & \beta_{24} & \beta_{15} \\ \beta_{03} & \beta_{13} & \beta_{04} & \beta_{23} & \beta_{14} & \beta_{05} & \beta_{33} & \beta_{24} & \beta_{15} & \beta_{06} \end{array} \right) \end{array}.$$

When the matrix $M(n)$ exhibits a column relation, it can be written as $p(X, Y) = \mathbf{0}$ for some polynomial $p(x, y) = \sum_{ij} a_{ij} x^i y^j \in \mathbb{R}[x, y]_n$, where each monomial is replaced by the column of $M(n)$ indexed with this monomial and $\mathbf{0}$ stands for a zero vector; this concept is known as functional calculus. For convenience, we sometimes refer to this column dependence relation as that of the moment sequence. Although column relations themselves are not polynomials, they can be interpreted as such and provide vital information about a representing measure for $M(n)$. Specifically, if β has a representing measure, the following condition must be satisfied [6]:

$$p(X, Y) = \mathbf{0} \implies (pq)(X, Y) = \mathbf{0} \text{ for every } q \in \mathbb{R}[x, y] \text{ with } \deg(pq) \leq n.$$

In this case, $M(n)$ is said to be **recursively generated**.

Let $\mathcal{Z}(p) := \{(x, y) \in \mathbb{R}^2 \colon p(x, y) = 0\}$ denote the zero set of the polynomial p. The following result helps determine the support of a representing measure for a truncated moment sequence:

Proposition 1.1 ([5, Proposition 3.1])**.** *Suppose μ is a representing measure for $\beta^{(2n)}$. For $p \in \mathbb{R}[x, y]_n$,*

$$\operatorname{supp} \mu \subseteq \mathcal{Z}(p) \iff p(X, Y) = \mathbf{0}.$$

Now, we aim to define a set contained in the support of representing measures; the **algebraic variety** of $\beta \equiv \beta^{(2n)}$ is given by

$$\mathcal{V}_\beta \equiv \mathcal{V}(M(n)) := \bigcap_{p(x,y)=\mathbf{0},\, \deg p \leq n} \mathcal{Z}(p). \tag{1}$$

Using Proposition 1.1 the following holds: if $M(n)$ admits a representing measure μ, then

$$\operatorname{supp} \mu \subseteq \mathcal{V}_\beta$$

and

$$\operatorname{rank} M(n) \leq \operatorname{card} \operatorname{supp} \mu \leq \operatorname{card} \mathcal{V}_\beta.$$

The inequality in the above is known as the **variety condition**.

Here, we introduce the **Riesz functional** defined on the space of all monomials. For a given moment sequence $\beta \equiv \beta^{(m)}$, the linear functional Λ_β is given by

$$\Lambda_\beta \left(\sum_{0 \leq i+j \leq m} a_{i,j} x^i y^j \right) := \sum_{0 \leq i+j \leq m} a_{i,j} \beta_{i,j},$$

where all $a_{i,j}$ are real numbers and $i, j \in \mathbb{Z}_+$. We say that Λ_β is **K-positive** for a closed set $K \subseteq \mathbb{R}^2$ if

$$\Lambda_\beta(p) \geq 0 \quad \text{for all } p \in \mathbb{R}[x,y]_m \text{ such that } p|_K \geq 0.$$

If, in addition, the conditions $p|_K \geq 0$ and $p|_K \not\equiv 0$ imply $\Lambda_\beta(p) > 0$, then Λ_β is said to be **strictly K-positive**. When $K = \mathbb{R}^2$, we simply refer to Λ_β as **positive**. The K-positivity of Λ_β is a necessary condition for β to admit a **K-representing measure**, that is, a representing measure supported in K. Conversely, Haviland proved (in [15] for $K = \mathbb{R}^n$ and in [16] for general $K \subseteq \mathbb{R}^n$) that K-positivity is also sufficient for the existence of a K-representing measure in the case of infinite moment sequences. A truncated analogue of Haviland's theorem can be found in [8], while a strictly K-positive version was established in [11] (see also [19, Theorem 1.30]).

One of the most significant results in truncated moment theory is the Flat Extension Theorem. This theorem states that if $M(n)$ has a rank-preserving positive extension $M(n+1)$, then $\beta^{(2n)}$ possesses a rank $M(n)$-atomic representing measure [5]. The extension $M(n+1)$ is referred to as a **flat extension**. A notable special case arises when $\operatorname{rank} M(n) = \operatorname{rank} M(n-1)$. In this situation, $M(n)$ is referred to as **flat**, and $\beta^{(2n)}$ has a unique rank $M(n)$-atomic representing measure.

We now briefly outline how to construct a flat extension of $M(n)$. Notice that each rectangular block of $M(n)$ with the same degree moments forms a Hankel matrix. To construct an extension $M(n+1)$, consider the following form:

$$M(n+1) = \begin{pmatrix} M(n) & B \\ B^T & C \end{pmatrix},$$

where B and C are Hankel matrices with some new moments of degree $2n-1$ and $2n$, respectively. To ensure that a prospective moment matrix $M(n+1)$ is positive semidefinite, we need the following classical result:

Theorem 1.2 ([1, 20])**.** *Let A, B, and C be matrices of real numbers, with A and C being square matrices. Then*

$$\begin{pmatrix} A & B \\ B^T & C \end{pmatrix} \succeq 0 \iff \begin{cases} A \succeq 0, \\ B = AW \text{ for some } W, \\ C \succeq W^T A W. \end{cases} \iff \begin{cases} A \succeq 0, \\ B = AW \text{ for some } W, \\ C \succeq B^T A^\dagger B, \end{cases}$$

where $A^\dagger$ stands for the Moore-Penrose inverse of A. Moreover,

$$\operatorname{rank} \begin{pmatrix} A & B \\ B^T & C \end{pmatrix} = \operatorname{rank} A \iff C = B^T A^\dagger B.$$

Although finding the positive extension may appear straightforward, verifying that the C-block has a moment structure is not trivial.

Recently, many intriguing interactions between moment theory and algebraic geometry have been uncovered. Solving truncated moment problems can be interpreted as finding the roots of a system of multivariate polynomial equations. By Richter's result [17] (see also [19, Theorem 1.24]), if a moment sequence $\beta^{(2n)}$ has one or more representing measures, then at least one of these measures must be finitely atomic. Consequently, if a real sequence $\beta^{(2n)}$ is associated with a finitely atomic representing measure μ, it can be expressed as

$$\mu = \sum_{\ell=1}^{r} \rho_\ell \delta_{(x_\ell, y_\ell)},$$

where $r \leq \dim \mathbb{R}[x,y]_{2n}$. Our task is to determine positive numbers $\rho_1, \ldots, \rho_r$ (referred to as *densities*) and points $(x_1, y_1), \ldots, (x_r, y_r)$ (referred to as *atoms*) of the measure μ, such that for $i, j \in \mathbb{Z}_+$ and $0 \leq i + j \leq 2n$,

$$\beta_{i,j} = \sum_{\ell=1}^{r} \rho_\ell x_\ell^i y_\ell^j$$

Affine Linear Transformation. The following discusses a method for simplifying the moment problem using invertible affine linear transformations, specifically the invariance of moment problems under degree-one transformations. The complex version of this approach is detailed in [7], and we adopt the same notation to develop its real counterpart.

For $a, b, c, d, e, f \in \mathbb{R}$ with $bf \neq ce$, define

$$\Psi(x,y) \equiv (\Psi_1(x,y), \Psi_2(x,y)) := (a + bx + cy, d + ex + fy) \quad \text{for } x, y \in \mathbb{R}.$$

If Λ_β represents the Riesz functional associated with a given $\beta \equiv \beta^{(2n)}$, then we can construct a new equivalent moment sequence $\tilde{\beta}^{(2n)}$ with $\tilde{\beta}_{i,j} := \Lambda_\beta(\Psi_1^i \Psi_2^j)$ for $i, j \in \mathbb{Z}_+$ and $0 \leq i + j \leq 2n$. It follows that $\Lambda_{\tilde{\beta}}(p) = \Lambda_\beta(p \circ \Psi)$ for all $p \in \mathbb{R}[x,y]_n$. For more details, refer to [7].

Truncated Moment Sequences with an Infinite Algebraic Variety. The moment matrix $M(n)(\beta^{(2n)})$ (or the moment sequence) is said to be **p-pure** if its only column relations are those recursively derived from a polynomial $p \in \mathbb{R}[x,y]_n$. Thus, $\mathcal{V}_\beta$ is precisely $\mathcal{Z}(p)$; in other words, the algebraic variety of β is infinite. When the algebraic variety associated with a truncated moment sequence is finite, a clear procedure exists for solving the moment problem [13]. However, concrete solutions for $M(n)$ with $n \geq 3$ are scarce and challenging to study [14, 21, 22, 24, 25, 26, 27].

Carathéodory number. In addition to solvability, we study the *Carathéodory number* of a truncated moment sequence, i.e., the minimal number of atoms in a representing measure. Recently, the Carathéodory number of real plane cubics with smooth projectivization was investigated in [2] using tools from algebraic geometry. In particular, for degree $2n$ sequences with a p-pure moment matrix it is shown that the Carathéodory number is at most $3n + 1$, and the cases where it equals $3n$ are characterized in terms of the number of connected components of $\mathcal{Z}(p)$ (see [2, Section 7]). Note that the cubic curve studied in Section 3 below does *not* satisfy the projective smoothness hypothesis, so our determination of the Carathéodory number in Theorem 3.1 does not follow from [2]. We also note that asymptotic estimates for Carathéodory numbers on affine plane curves were studied recently in [10, 18].

Main Results. In this paper we focus on the TMP on curves of the form

$$xy = q_m x^m + q(x) + \alpha y, \quad \text{where } q_m \in \mathbb{R} \setminus \{0\},\ q(x) \in \mathbb{R}[x]_{m-1},\ \alpha \in \mathbb{R}.$$

By applying the affine linear transformation $(x, y) \mapsto (x + \alpha, q_m y)$, it suffices to solve the TMP on curves with the simpler form

$$xy = x^m + r(x), \quad \text{where } r(x) \in \mathbb{R}[x]_{m-1},\ r(0) \neq 0. \tag{2}$$

These curves have a parametrization $(x(t), y(t)) = (t, t^{m-1} + \frac{r(t)}{t})$, $t \in \mathbb{R}$, $t \neq 0$. The TMP on any rational curve is equivalent to a univariate TMP, where

some moments are missing and the measure must vanish at certain points. This observation simplifies solving the original TMP, since univariate TMPs are easier to tackle and related technique was already exploited in [22, 24, 25, 26, 27]. To address the TMP on (2), the solution to the strong Hamburger TMP [26] is required, i.e., the $\mathbb{R}$–representing measure of the univariate sequence must vanish in $\{0\}$. The original motivation for this paper was the cubic relation of type (2) with $m = 3$, since after applying an affine linear transformation, every irreducible cubic relation has one of four canonical forms, with type (2) and $m = 3$ being one of them. The other three types are $y = q(x)$, $y^2 = q(x)$ and $xy^2 + ay = q(x)$ for some $q \in \mathbb{R}[x]_3$ and $a \in \mathbb{R}$. A concrete solution to $y = q(x)$ is known [14], while concrete solutions to the other two types are known for $y^2 = x^3$ [24] and $xy^2 = 1$ [26].

Our first main result (see Theorem 2.2) is the solution to the TMP on (2) in terms of the corresponding univariate TMP with gaps. The solution is a Hankel positive semidefinite (psd) completion problem, i.e., the question is under what conditions the missing anti-diagonals of a partially defined Hankel matrix can be chosen so that the completion is psd and satisties two additional constraints, coming from the solution to the strong Hamburger TMP. Our second main result (see Theorems 3.1 and 3.2) solves this completion problem corresponding to the cubic case ($m = 3$ in (2)) in terms of concrete numerical conditions. We also determine the minimal number of atoms in a representing measure and illustrate the result with a numerical example (see Example 3.3). Our third main result (see Theorems 4.1 and 4.2) solves the completion problem for the quartic case ($m = 4$ in (2)) in terms of feasibility of a three-variable system of inequalities.

2. TMP ON $xy = q_m x^m + \sum_{s=0}^{m-1} q_s x^s$ WITH $q_0 \neq 0$

In this section we prove that the $\mathcal{Z}(p)$–TMP for

$$p(x, y) = xy - q_m x^m - \sum_{s=0}^{m-1} q_s x^s,$$

where each $q_s \in \mathbb{R}$ and $q_0 \neq 0$, $q_m \neq 0$, is equivalent to the Hankel positive semidefinite completion problem (see Theorem 2.2).

Let $\beta \equiv \left\{\beta_{i,j}^{(2n)}\right\}$ be a sequence with for $i, j \in \mathbb{Z}_+$, $i + j \leq 2n$. Let p be as in the first paragraph. By using an affine linear transformation, we may assume that $q_m = 1$. We need to convert the given bivariate moment sequence into its equivalent univariate counterpart. First, let us reorder the univariate indices

$$-2n, -2n + 1, \ldots, -1, 0, 1, \ldots, 2(m - 1)n - 1, 2(m - 1)n$$

in the following way (Refer to Remark 2.1 for the rationale behind this transformation.):

$$\begin{aligned} &\text{row } 0: \quad 0, 1, \ldots, 2n, \\ &\text{row } k: \quad -k, (2n - k + 1) + (m - 1)(k - 1) + 1, \ldots, (2n - k) + (m - 1)k, \end{aligned} \tag{3}$$

for $k = 1, \ldots, 2n$. Now we adapt a notation $h(k)$ for row k ($k = 1, \ldots, 2n$) in the following way:

$$\text{row } k: \quad -k, h(k), h(k) + 1, \ldots, (2n - k) + (m - 1)k,$$

where

$$h(k) := \max\{2n - k + 1 + (m - 1)(k - 1), (m - 1)k\} + 1. \tag{4}$$

Remark 2.1. The reason for this adaptation is the fact that expressing y from the equality $p(x,y)=0$ and then raising to the power of the index of the row the relation will be of the form

$$y^k = \Big(\sum_{s=0}^{m} q_s x^{s-1}\Big)^k = q_m^k x^{(m-1)k} + \sum_{i=-k}^{(m-1)k-1} r_i x^i$$

for some $r_i \in \mathbb{R}$. Multiplying this relation with $x, \ldots, x^{2n-k}$, we can successively express $x^{(m-1)k+j}$, $j = 1, \ldots, 2n-k$, out of the equations obtained. If $(m-1)k$ is larger than $(2n-k+1)+(m-1)(k-1)$, then some powers of x will be missing in this procedure. The missing powers will be precisely $(2n-k+1)+(m-1)(k-1)+1, \ldots, h(k)-1$.

As explained in Remark 2.1, the adaptation may result in the loss of some indices in each row. Let $\mathcal{I}$ be the set of indices remaining in the sequence after this adaptation. We define a map f on $\mathcal{I}$ by the rule

$$\begin{aligned} f(s) &\equiv (f_1(s), f_2(s)) \\ &:= \begin{cases} \big(s - \#(s)\cdot(m-1), \#(s)\big), & \text{if } s \geq 0, \\ \big(0, \#(s)\big), & \text{if } s < 0, \end{cases} \end{aligned} \tag{5}$$

where the index s is contained in row $\#(s)$.

Expressing y from the relation $p(x,y)=0$, we see that for $i, j \in \mathbb{Z}_+$ we have

$$\begin{aligned} & x^i \left(\sum_{s=0}^{m} q_s x^{s-1}\right)^j \\ &\quad = x^i \left(\sum_{\substack{k_0+\ldots+k_m=j,\\ k_0,\ldots,k_m\in\mathbb{Z}_+}} \frac{j!}{k_0!\cdots k_m!} q_m^{k_0} q_{m-1}^{k_1} \cdots q_0^{k_m} x^{\sum_{s=0}^m (m-1-s)k_s} \right) \\ &\quad = \sum_{\substack{k_0+\ldots+k_m=j,\\ k_0,\ldots,k_m\in\mathbb{Z}_+}} \frac{j!}{k_0!\cdots k_m!} q_m^{k_0} q_{m-1}^{k_1} \cdots q_0^{k_m} x^{\sum_{s=0}^m (m-1-s)k_s + i} \\ &\quad = \sum_{t=-j}^{(m-1)j} q_{j,t} x^{t+i}, \end{aligned} \tag{6}$$

where

$$q_{j,t} := \sum_{\substack{k_0+\ldots+k_m=j,\\ k_0,\ldots,k_m\in\mathbb{Z}_+,\\ \sum_{s=0}^m (m-1-s)k_s = t}} \frac{j!}{k_0!\cdots k_m!} q_m^{k_0} q_{m-1}^{k_1} \cdots q_0^{k_m}$$

Note that

$$q_{j,(m-1)j} = q_m^j = 1 \quad \text{and} \quad q_{j,-j} = q_0^j.$$

Hence,

$$x^{(m-1)j+i} = x^i \left(\sum_{s=0}^{m} q_s x^{s-1}\right)^j - \sum_{t=-j}^{(m-1)j-1} q_{j,t} x^{t+i}. \tag{7}$$

and

$$x^{i-j} = \frac{1}{q_0^j}\left[x^i\left(\sum_{s=0}^{m} q_s x^{s-1}\right)^j - \sum_{t=-j+1}^{(m-1)j} q_{j,t}x^{t+i}\right]. \tag{8}$$

Using the relations (6)–(8) we define a number γ_s for every $s \in \mathcal{I}$ following the order (3) by

$$\gamma_s := \begin{cases} \beta_{f(s)} - \displaystyle\sum_{t=-f_2(s)}^{(m-1)f_2(s)-1} q_{f_2(s),t}\gamma_{t+f_1(s)}, & \text{if } s \geq 0, \\ \dfrac{1}{q_0^{f_2(s)}}\left(\beta_{f(s)} - \displaystyle\sum_{t=-f_2(s)+1}^{(m-1)f_2(s)} q_{f_2(s),t}\gamma_t\right), & \text{if } s < 0. \end{cases} \tag{9}$$

Namely, we define γ_s in the order

$$\underbrace{\gamma_0, \gamma_1, \ldots, \gamma_{2n}}_{j=0}, \underbrace{\gamma_{-1}, \gamma_{h(1)}, \gamma_{h(1)+1} \cdots, \gamma_{2n+m-2}}_{j=1}, \cdots, \underbrace{\gamma_{-2n+1}, \gamma_{h(2n-1)}}_{j=2n-1}, \underbrace{\gamma_{-2n}}_{j=2n}. \tag{10}$$

If the moment γ_s does not belong to the index set $\mathcal{I}$, we define it as a **free moment**. Conversely, if $s \in \mathcal{I}$ but the expression for γ_s involves other free moments, then γ_s is not uniquely determined; in this case, we refer to γ_s as an **auxiliary moment**. If γ_s is not free or auxiliary, then it is called a **fully–determined moment**.

Let $k \in \mathbb{N}$. For $v = (v_0, \ldots, v_{2k}) \in \mathbb{R}^{2k+1}$ we define the corresponding Hankel matrix as

$$A_v := (v_{i+j})_{i,j=0}^{k} = \begin{pmatrix} v_0 & v_1 & v_2 & \cdots & v_k \\ v_1 & v_2 & \iddots & \iddots & v_{k+1} \\ v_2 & \iddots & \iddots & \iddots & \vdots \\ \vdots & \iddots & \iddots & \iddots & v_{2k-1} \\ v_k & v_{k+1} & \cdots & v_{2k-1} & v_{2k} \end{pmatrix}.$$

The main result of this section is the following solution to the $\mathcal{Z}(p)$–TMP for β.

Theorem 2.2. *Let $p(x,y) = xy - \sum_{s=0}^{m} q_s x^s$ with $q_i \in \mathbb{R}$, $q_0 \neq 0$, $q_m = 1$. Given a sequence $\beta \equiv \left\{\beta_{i,j}^{(2n)}\right\}$ for $i, j \in \mathbb{Z}_+$, $i + j \leq 2n$, let*

$$\gamma \equiv \gamma^{(-2n,2(m-1)n)} = (\gamma_{-2n}, \gamma_{-2n+1}, \ldots, \gamma_{-1}, \gamma_0, \gamma_1, \ldots, \gamma_{2(m-1)n})$$

be defined by the procedure above and suppose $\gamma_{j_1}, \ldots, \gamma_{j_p}$ are free moments. Then the following are equivalent:

- *(i) β admits a $\mathcal{Z}(p)$–representing measure.*
- *(ii) γ has a $\mathbb{R}$–representing measure μ for some choice of real values of free moments such that $\mu(\{0\}) = 0$.*
- *(iii) There is a choice of real values of free moments such that A_γ is positive semidefinite and one of the following holds:*
  - *(a) A_γ is positive definite.*
  - *(b) $\operatorname{rank} A_\gamma = \operatorname{rank} A_{\gamma^{(-2n,2(m-1)n-2)}} = \operatorname{rank} A_{\gamma^{(-2n+2,2(m-1)n)}}$.*

Proof. The equivalence $(ii) \Leftrightarrow (iii)$ is [26, Theorem 3.1]. It remains to prove the equivalence $(i) \Leftrightarrow (ii)$. By [17] (see also [19, Theorem 1.24]), it suffices to prove $(i) \Leftrightarrow (ii)$ for finitely atomic measures, and hence it is enough to establish the following claim.

Claim. Let $r \in \mathbb{N}$. A sequence γ admits a r–atomic $\mathbb{R}$–representing measure vanishing on $\{0\}$ if and only if β admits a r–atomic $\mathcal{Z}(p)$–representing measure.

Proof of the Claim. First we prove the forward implication. Let $\mu_x = \sum_{\ell=1}^{r} \rho_\ell \delta_{x_\ell}$ be a $\mathbb{R}$–representing measure for γ where $x_\ell \in \mathbb{R}\setminus\{0\}$ and $\rho_\ell > 0$ for each ℓ. We will prove that $\mu = \sum_{\ell=1}^{r} \rho_\ell \delta_{(x_\ell, y_\ell)}$, where $y_\ell = \sum_{s=0}^{m} q_s x_\ell^{s-1}$, is a $\mathcal{Z}(p)$–representing measure for β. We use induction on the index j in $\beta_{i,j}$, where $i + j \leq 2n$:

Base of induction: For $j = 0$, we see that

$$\beta_{i,0} = \gamma_i = \sum_{\ell=1}^{r} \rho_\ell x_\ell^i = \sum_{\ell=1}^{r} \rho_\ell x_\ell^i y_\ell^0,$$

where we used (9) and substituted $\gamma_i = \int x^i d\mu_x$.

Induction step: Assume that Claim holds for every $j \leq j_0 - 1$ for some $1 \leq j_0 \leq 2n$. Let us prove its validity for j_0. We consider two cases separately.

Case 1: (i, j_0) *is in the image of* f. Let $s = f^{-1}((i, j_0))$. Then we have

$$\beta_{i,j_0} = \sum_{t=-j_0}^{(m-1)j_0} q_{j_0,t}\, \gamma_{t+i},$$

where we used (9). Substituting $\gamma_{t+i} = \int x^{t+i}\, d\mu_x$, we obtain

$$\begin{aligned}
\beta_{i,j_0} &= \sum_{t=-j_0}^{(m-1)j_0} q_{j_0,t} \left(\sum_{\ell=1}^{r} \rho_\ell x_\ell^{t+i} \right) \\
&= \sum_{\ell=1}^{r} \rho_\ell \sum_{t=-j_0}^{(m-1)j_0} q_{j_0,t}\, x_\ell^{t+i} \\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^i \sum_{t=-j_0}^{(m-1)j_0} q_{j_0,t}\, x_\ell^{t} \\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^i y_\ell^{j_0},
\end{aligned}$$

where in the last step we used (6).

Case 2: (i, j_0) *is not in the image of* f. Since (i, j_0) is not in the image of f, we have

$$i \neq 0 \quad \text{and} \quad i + (m-1)j_0 \leq (2n - j_0 + 1) + (m-1)(j_0 - 1). \tag{11}$$

The first condition in (11) is immediate, since $f(-j) = (0, j)$ for every $0 \leq j \leq 2n$. If (i, j_0) were in the image of f, then $f^{-1}((i, j_0)) = i + (m-1)j_0$. The second

inequality in (11) is equivalent to

$$i \leq -m + 2n - j_0 + 2. \tag{12}$$

Since the moment sequence is recursively generated, we must have

$$\beta_{i,j_0} = \sum_{s=0}^{m} q_s \, \beta_{i-1+s,\, j_0-1}. \tag{13}$$

Observe that $i - 1 + s \geq 0$ for each s, $j_0 - 1 \geq 0$, and

$$\begin{aligned} i + s + j_0 - 2 &\leq i + m + j_0 - 2 && \text{(since } s \leq m\text{)} \\ &\leq -m + 2n - j_0 + 2 + m + j_0 - 2 && \text{(by (12))} \\ &= 2n. \end{aligned}$$

It follows that each $\beta_{i-1+s,\, j_0-1}$ appearing in (13) belongs to the original sequence. We now compute

$$\begin{aligned} \beta_{i,j_0} &= \sum_{s=0}^{m} q_s \, \beta_{i-1+s,\, j_0-1} && \text{(by (13))} \\ &= \sum_{s=0}^{m} q_s \left(\sum_{\ell=1}^{r} \rho_\ell x_\ell^{i-1+s} y_\ell^{j_0-1} \right) && \text{(by the induction hypothesis)} \\ &= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i} y_\ell^{j_0-1} \left(\sum_{s=0}^{m} q_s x_\ell^{s-1} \right) \\ &= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i} y_\ell^{j_0-1} y_\ell && \text{(by (6) with } i = 0,\ j = 1\text{)} \\ &= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i} y_\ell^{j_0}, \end{aligned}$$

This completes the induction step and proves the forward implication.

It remains to prove the backward implication of Claim. Let

$$\mu = \sum_{\ell=1}^{r} \rho_\ell \delta_{(x_\ell, y_\ell)}$$

be a $\mathcal{Z}(p)$-representing measure for β, where $(x_\ell, y_\ell) \in \mathcal{Z}(p)$ and $\rho_\ell > 0$ for each ℓ. We will show that $\mu_x = \sum_{\ell=1}^{r} \rho_\ell \delta_{x_\ell}$ is a representing measure for γ, which by construction vanishes on $\{0\}$ (since $x_\ell \neq 0$ for all ℓ). We proceed by induction on the index i in γ_i, ordered according to (3). For $i = 0$, we have

$$\gamma_0 = \beta_{0,0} = \sum_{\ell=1}^{r} \rho_\ell x_\ell^0,$$

so the claim holds. Assume it holds up to some index s_0 and prove it for s_1. We distinguish two cases.

Case 1: *γ_{s_1} is a free moment.*

In this case, we define

$$\gamma_{s_1} = \sum_{\ell=1}^{r} \rho_\ell x_\ell^{s_1}.$$

Case 2: γ_{s_1} *is not a free moment.*

In this case, γ_{s_1} is either a fully determined or an auxiliary moment; in either situation, we have $s_1 \in \mathcal{I}$. Write

$$f(s_1) = (i_1, j_1).$$

Subcase 2(a): $s_1 \geq 0$. We have

$$\begin{aligned}
\gamma_{s_1} &= \beta_{i_1,j_1} - \sum_{t=-j_1}^{(m-1)j_1-1} q_{j_1,t}\,\gamma_{t+i_1} && \text{(by (9))}\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i_1} y_\ell^{j_1} - \sum_{t=-j_1}^{(m-1)j_1-1} q_{j_1,t} \left(\sum_{\ell=1}^{r} \rho_\ell x_\ell^{t+i_1} \right) && \text{(by the induction hypo.)}\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i_1} \left(y_\ell^{j_1} - \sum_{t=-j_1}^{(m-1)j_1-1} q_{j_1,t} x_\ell^{t} \right)\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{i_1} x_\ell^{(m-1)j_1} && \text{(by (6))}\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{(m-1)j_1+i_1} && \text{(by the definition of } (i_1, j_1))\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{s_1}.
\end{aligned}$$

Subcase 2(b): $s_1 < 0$. In this case, $i_1 = 0$, and

$$\begin{aligned}
\gamma_{s_1} &= \frac{1}{q_0^{j_1}} \left(\beta_{0,j_1} - \sum_{t=-j_1+1}^{(m-1)j_1} q_{j_1,t}\,\gamma_t \right) && \text{(by (9))}\\
&= \frac{1}{q_0^{j_1}} \left[\sum_{\ell=1}^{r} \rho_\ell y_\ell^{j_1} - \sum_{t=-j_1+1}^{(m-1)j_1} q_{j_1,t} \left(\sum_{\ell=1}^{r} \rho_\ell x_\ell^{t} \right) \right] && \text{(by the induction hypo.)}\\
&= \sum_{\ell=1}^{r} \rho_\ell \frac{1}{q_0^{j_1}} \left(y_\ell^{j_1} - \sum_{t=-j_1+1}^{(m-1)j_1} q_{j_1,t} x_\ell^{t} \right) && \text{(by (6))}\\
&= \sum_{\ell=1}^{r} \rho_\ell x_\ell^{-j_1},
\end{aligned}$$

This completes the proof of the backward implication of the claim, and hence the proof of the equivalence between (i) and (ii) is established. □

3. Concrete solution to the TMP on $p(x, y) = xy - x^3 - \sum_{i=0}^{2} q_i x^i$, $q_0 \neq 0$

In this section we derive explicit numerical conditions, in the case $m = 3$, for the existence of free moments in Theorem 2.2, and thereby obtain an explicit solution to the TMP. We briefly comment on the standing assumption $q_0 \neq 0$: if $q_0 = 0$, then the column relation induced by $p(X, Y) = \mathbf{0}$ becomes reducible, and the associated TMP can be handled by existing results. We also determine the

minimum number of atoms in a representing measure in this setting. Whenever a representing measure exists, one can be chosen with at most rank $M(n)+1$ atoms. Theorems 3.1, treating the p-pure case, and Theorem 3.2, addressing the singular case, characterize when the additional atom is necessary.

Assume the notation from Section 2 and let $\gamma \equiv \gamma^{(-2n,4n)}$ be defined by (9) for $m = 3$. Recalling (4), we have

$$h(k) = \max\{2n+k-1, 2k\}+1 = \begin{cases} 2n+k, & \text{if } k < 2n, \\ 4n+1, & \text{if } k = 2n. \end{cases}$$

Hence, $\mathcal{I} = \{-2n, -2n+1, \ldots, 4n-1\}$ and the only free moment is γ_{4n}. We define γ_s in the order

$$\underbrace{\gamma_0, \gamma_1, \ldots, \gamma_{2n}}_{j=0}, \underbrace{\gamma_{-1}, \gamma_{2n+1}}_{j=1}, \ldots, \underbrace{\gamma_{-k}, \gamma_{2n+k}}_{j=k}, \ldots, \underbrace{\gamma_{-2n+1}, \gamma_{4n-1}}_{j=2n-1}, \underbrace{\gamma_{-2n}}_{j=2n} \tag{14}$$

by (9). The only auxiliary moment is γ_{-2n}; except for γ_{4n} and γ_{-2n}, all remaining moments are fully determined. More explicitly, for $j = 0$ we have

$$\gamma_0 = \beta_{0,0}, \quad \gamma_1 = \beta_{1,0}, \quad \ldots, \quad \gamma_{2n} = \beta_{2n,0};$$

for $j = 1$,

$$\gamma_{-1} = \frac{1}{q_0}\Big(\beta_{0,1} - \sum_{t=0}^{2} q_{1,t}\gamma_t\Big),$$
$$\gamma_{2n+1} = \beta_{2n-1,1} - \sum_{t=-1}^{1} q_{1,t}\gamma_{t+2n-1};$$
$$\vdots$$

for $j = k$,

$$\gamma_{-k} = \frac{1}{q_0^k}\Big(\beta_{0,k} - \sum_{t=-k+1}^{2k} q_{k,t}\gamma_t\Big),$$
$$\gamma_{2n+k} = \beta_{2n-k,k} - \sum_{t=-k}^{2k-1} q_{k,t}\gamma_{t+2n-k};$$
$$\vdots$$

for $j = 2n-1$,

$$\gamma_{-2n+1} = \frac{1}{q_0^{2n-1}}\Big(\beta_{0,2n-1} - \sum_{t=-2n+2}^{4n-2} q_{2n-1,t}\gamma_t\Big),$$
$$\gamma_{4n-1} = \beta_{1,2n-1} - \sum_{t=-2n+1}^{4n-3} q_{2n-1,t}\gamma_{t+1};$$

for $j = 2n$,

$$\gamma_{-2n}(\gamma_{4n}) = \frac{1}{q_0^{2n}}\Big(\beta_{0,2n} - \sum_{t=-2n+1}^{4n-1} q_{2n,t}\gamma_t - \gamma_{4n}\Big) =: D - q_0^{-2n}\gamma_{4n}.$$

We introduce a new variable t for γ_{4n} and let $A_{\gamma(t)}$ denote the corresponding Hankel matrix:

$$A_{\gamma(t)} = \begin{array}{c} \\ T^{-n} \\ T^{-n+1} \\ \vdots \\ T^{-1} \\ 1 \\ T \\ \vdots \\ T^{2n} \end{array} \begin{array}{c} \begin{array}{cccccccc} T^{-n} & T^{-n+1} & \cdots & T^{-1} & 1 & T & \cdots & T^{2n} \end{array} \\ \begin{pmatrix} \gamma_{-2n}(t) & \gamma_{-2n+1} & \cdots & \gamma_{-n-1} & \gamma_{-n} & \gamma_{-n+1} & \cdots & \gamma_n \\ \gamma_{-2n+1} & \gamma_{-2n+2} & \cdots & \gamma_{-n} & \gamma_{-n+1} & \gamma_{-n+2} & \cdots & \gamma_{n+1} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \gamma_{-n-1} & \gamma_{-n} & \cdots & \gamma_{-2} & \gamma_{-1} & \gamma_0 & \cdots & \gamma_{2n-1} \\ \gamma_{-n} & \gamma_{-n+1} & \cdots & \gamma_{-1} & \gamma_0 & \gamma_1 & \cdots & \gamma_{2n} \\ \gamma_{-n+1} & \gamma_{-n+2} & \cdots & \gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{2n+1} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \gamma_n & \gamma_{n+1} & \cdots & \gamma_{2n-1} & \gamma_{2n} & \gamma_{2n+1} & \cdots & t \end{pmatrix} \end{array}.$$

For $i \le j$ we set

$$\vec{T}^{(i,j)} := \begin{pmatrix} T^i & T^{i+1} & \cdots & T^j \end{pmatrix}.$$

With this notation, $A_{\gamma(t)}$ admits the block decomposition

$$A_{\gamma(t)} = \begin{array}{c} \\ T^{-n} \\ (\vec{T}^{(-n+1,2n-1)})^T \\ T^{2n} \end{array} \begin{array}{c} \begin{array}{ccc} T^{-n} & \vec{T}^{(-n+1,2n-1)} & T^{2n} \end{array} \\ \begin{pmatrix} D - q_0^{-2n}t & b^T & \gamma_n \\ b & A_{\widetilde{\gamma}} & c \\ \gamma_n & c^T & t \end{pmatrix} \end{array},$$

where

$$\begin{aligned} b^T &= \begin{pmatrix} \gamma_{-2n+1} & \cdots & \gamma_{-n-1} & \gamma_{-n} & \gamma_{-n+1} & \cdots & \gamma_{n-1} \end{pmatrix}, \\ c^T &= \begin{pmatrix} \gamma_{n+1} & \cdots & \gamma_{2n-1} & \gamma_{2n} & \gamma_{2n+1} & \cdots & \gamma_{4n-1} \end{pmatrix}, \\ \widetilde{\gamma} &= (\gamma_{-2n+2}, \gamma_{-2n+3}, \ldots, \gamma_{4n-2}). \end{aligned}$$

The following theorem provides an explicit solution to the p-pure TMP.

Theorem 3.1 (Pure case)**.** *Let $p(x,y) = xy - \sum_{s=0}^{3} q_s x^s$, where $q_i \in \mathbb{R}$, $q_0 \neq 0$, and $q_3 = 1$. Given a p-pure sequence $\beta \equiv \left\{\beta_{i,j}^{(2n)}\right\}$ for $i, j \in \mathbb{Z}_+$, $i + j \le 2n$, let*

$$\gamma(t) = (\gamma_{-2n}(t), \gamma_{-2n+1}, \ldots, \gamma_{-1}, \gamma_0, \gamma_1, \ldots, \gamma_{4n-1}, t)$$

be defined by the procedure above. Assume the notation introduced above and define

$$\begin{aligned} t_{\min} &:= c^T A_{\widetilde{\gamma}}^{-1} c, \\ t_{\max} &:= q_0^{2n}(D - b^T A_{\widetilde{\gamma}}^{-1} b), \\ E &:= b^T A_{\widetilde{\gamma}}^{-1} c c^T A_{\widetilde{\gamma}}^{-1} b - 2 b^T A_{\widetilde{\gamma}}^{-1} c \gamma_n + \gamma_n^2. \end{aligned} \tag{15}$$

Then the following are equivalent:

(i) β admits a representing measure.
(ii) β admits a representing measure with at most $3n+1$ atoms.
(iii) $t_{\min} < t_{\max}$ and $(t_{\max} - t_{\min})^2 \ge 4q_0^{2n}E$.

Moreover, whenever these conditions hold, the minimum number of atoms in a representing measure is $3n$ if $E > 0$, and $3n+1$ if $E = 0$.

The next theorem provides a solution of the singular $\mathcal{Z}(p)$–TMP when the associated algebraic variety is finite.

Theorem 3.2 (Singular case)**.** *Let $p(x,y) = xy - \sum_{s=0}^{3} q_s x^s$, where $q_i \in \mathbb{R}$, $q_0 \neq 0$, and $q_3 = 1$. Given a sequence $\beta \equiv \left\{\beta_{i,j}^{(2n)}\right\}$ for $i,j \in \mathbb{Z}_+$, $i+j \leq 2n$, with* rank $M(n) < 3n$*, where $n \geq 3$ and $M(n) \succeq 0$ is recursively generated with $p(X,Y) = \mathbf{0}$, let*

$$\gamma(t) = (\gamma_{-2n}(t), \gamma_{-2n+1}, \ldots, \gamma_{-1}, \gamma_0, \gamma_1, \ldots, \gamma_{4n-1}, t)$$

be defined by the procedure above. Assuming the notation introduced above, the following are equivalent:

- *(i) β admits a representing measure.*
- *(ii) β admits a representing measure with at most* (rank $M(n)+1$) *atoms.*
- *(iii) (a) $c = A_{\widetilde{\gamma}} w$ for some $w \in \mathbb{R}^{3n-1}$ and* rank $A_{\gamma(t_0)}$ = rank $A_{\widetilde{\gamma}}$ *for $t_0 := c^T A_{\widetilde{\gamma}}^{\dagger} c$, where $A_{\widetilde{\gamma}}^{\dagger}$ denotes the Moore-Penrose inverse of $A_{\widetilde{\gamma}}$. Or (b) $A_{\widetilde{\gamma}} \succ 0$ and $t_{\min} < t_{\max}$ with $t_{\min}, t_{\max}$ as in* (15)*.*

Moreover, the minimum number of atoms is $r :=$ rank $M(n)$ *in case (iii)(a), and $r+1 = 3n$ in case (iii)(b).*

Proof of Theorem 3.1. Before proving the equivalences in the theorem, we establish several auxiliary claims. For integers $i \leq j$, we denote by $(A_{\gamma(t)})\big|_{\vec{T}^{(i,j)}}$ the principal submatrix of $A_{\gamma(t)}$ on rows and columns labelled by elements from $\vec{T}^{(i,j)}$.

Claim 1. $A_1 := (A_{\gamma(t)})\big|_{\vec{T}^{(-n,2n-1)}} \succeq 0 \iff t \leq t_{\max}$.

Proof of Claim 1. By the block form of $A_{\gamma(t)}$,

$$A_1 = \begin{array}{c} \\ T^{-n} \\ (\vec{T}^{(-n+1,2n-1)})^T \end{array} \begin{array}{c} \begin{array}{cc} T^{-n} & \vec{T}^{(-n+1,2n-1)} \end{array} \\ \begin{pmatrix} D - q_0^{-2n} t & b^T \\ b & A_{\widetilde{\gamma}} \end{pmatrix} \end{array}.$$

Since $A_{\widetilde{\gamma}}$ is positive definite, Theorem 1.2 yields:

$$A_1 \succeq 0 \quad \Longleftrightarrow \quad D - q_0^{-2n} t \geq b^T A_{\widetilde{\gamma}}^{-1} b \quad \Longleftrightarrow \quad t \leq t_{\max},$$

which proves Claim 1. ∎

Claim 2. $A_2 := (A_{\gamma(t)})\big|_{\vec{T}^{(-n+1,2n)}} \succeq 0 \iff t_{\min} \leq t$.

Proof of Claim 2. Again using the block form,

$$A_2 = \begin{array}{c} \\ (\vec{T}^{(-n+1,2n-1)})^T \\ T^{2n} \end{array} \begin{array}{c} \begin{array}{cc} \vec{T}^{(-n+1,2n-1)} & T^{2n} \end{array} \\ \begin{pmatrix} A_{\widetilde{\gamma}} & c \\ c^T & t \end{pmatrix} \end{array}.$$

Since $A_{\widetilde{\gamma}}$ is positive definite, Theorem 1.2 implies the following:

$$A_2 \succeq 0 \quad \Longleftrightarrow \quad t \geq c^T A_{\widetilde{\gamma}}^{-1} c = t_{\min},$$

which proves Claim 2. ∎

Claim 3. Let $t = t_{\min} + w$ with $w > 0$, and define

$$Q(w) := -\frac{w^2}{q_0^{2n}} + \frac{t_{\max} - t_{\min}}{q_0^{2n}}\, w - E. \tag{16}$$

Then

$$A_{\gamma(t)} \succeq 0 \quad \Longleftrightarrow \quad \big(Q(w) \geq 0 \text{ and } 0 < w \leq t_{\max} - t_{\min}\big), \tag{17}$$

$$\Longrightarrow \quad (t_{\max} - t_{\min})^2 \geq 4q_0^{2n} E. \tag{18}$$

Proof of Claim 3. If $A_{\gamma(t)} \succeq 0$, then necessarily $A_1 \succeq 0$ and $A_2 \succeq 0$. By Claims 1 and 2 this forces $t \in [t_{\min}, t_{\max}]$, i.e.

$$0 < w \leq t_{\max} - t_{\min}.$$

Moreover, since $w > 0$ we have $t > t_{\min}$, hence $A_2 \succ 0$. Using the block inversion formula [29, Formula (0.7.2)], we obtain

$$A_2^{-1} = \begin{pmatrix} A_{\widetilde{\gamma}}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{w} \begin{pmatrix} A_{\widetilde{\gamma}}^{-1} c c^t A_{\widetilde{\gamma}}^{-1} & -A_{\widetilde{\gamma}}^{-1} c \\ -c^T A_{\widetilde{\gamma}}^{-1} & 1 \end{pmatrix}.$$

Applying Theorem 1.2 to the decomposition of $A_{\gamma(t)}$ with respect to the block A_2, we have

$$A_{\gamma(t)} \succeq 0 \Longleftrightarrow D - \frac{t}{q_0^{2n}} \;\geq\; \begin{pmatrix} b^T & \gamma_n \end{pmatrix} A_2^{-1} \begin{pmatrix} b \\ \gamma_n \end{pmatrix}.$$

Substituting $t = t_{\min} + w$ and the expression for A_2^{-1} gives

$$\begin{aligned} A_{\gamma(t)} \succeq 0 \Longleftrightarrow \; & -\frac{w}{q_0^{2n}} + \Big(D - b^T A_{\widetilde{\gamma}}^{-1} b - \frac{t_{\min}}{q_0^{2n}}\Big) \\ & - \frac{1}{w}\Big(b^T A_{\widetilde{\gamma}}^{-1} c c^T A_{\widetilde{\gamma}}^{-1} b - 2\, b^T A_{\widetilde{\gamma}}^{-1} c\, \gamma_n + \gamma_n^2\Big) \geq 0. \end{aligned}$$

Multiplying by $w > 0$ and using the definitions of $t_{\max}$ and E yields precisely $Q(w) \geq 0$, which proves (17). Finally, Q is a concave quadratic polynomial, whose maximum is attained at $w_0 = (t_{\max} - t_{\min})/2$ as

$$Q(w_0) = \frac{(t_{\max} - t_{\min})^2}{4q_0^{2n}} - E.$$

Thus, $Q(w) \geq 0$ implies $Q(w_0) \geq 0$, i.e.,

$$(t_{\max} - t_{\min})^2 \geq 4q_0^{2n} E,$$

which proves (18). ∎

We now prove $(i) \Rightarrow (iii)$. Assume that β admits a representing measure. By Theorem 2.2, there exists a choice of γ_{4n} (equivalently, $t \in \mathbb{R}$) such that $\gamma(t)$ admits a representing measure; in particular, $A_{\gamma(t)} \succeq 0$. Claims 1 and 2 then imply $t_{\min} \leq t_{\max}$. We next show that the inequality is strict. Suppose, to the contrary, that $t_{\min} = t_{\max}$. Then t is forced to equal this common value, implying that both A_1 and A_2 are singular. Equivalently, the first and last columns of $A_{\gamma(t)}$ lie in the span of the intermediate columns, so rank $A_{\gamma(t)} = 3n - 1$. By [26, Theorem 3.1], $\gamma(t)$ would then admit a $(3n-1)$–atomic representing measure, and hence so would β, contradicting the p-purity of β (which implies rank $M(n) = 3n$). Therefore $t_{\min} < t_{\max}$. The remaining inequality in (iii) follows from Claim 3 if $t > t_{\min}$; if

$t = t_{\min}$, positive semidefiniteness forces $\gamma_n = b^T A_{\widetilde{\gamma}}^{-1} c$, and hence $E = 0$.

Next we prove $(iii) \Rightarrow (ii)$. First suppose that $E > 0$. By [26, Theorem 3.1], it suffices to exhibit t such that

$$A_{\gamma(t)} \succeq 0 \quad \text{and} \quad \text{rank } A_{\gamma(t)} = \text{rank } A_1 = \text{rank } A_2 = 3n. \tag{19}$$

By Claims 1 and 2, we must have $t \in [t_{\min}, t_{\max}]$. If $t = t_{\min}$ (resp. $t = t_{\max}$), then A_2 (resp. A_1) is singular, so (19) cannot hold. Hence $t \in (t_{\min}, t_{\max})$, i.e. $w \in (0, t_{\max} - t_{\min})$. By assumption in (iii) and $E > 0$, there exists $w \in (0, t_{\max} - t_{\min})$ such that $Q(w) = 0$. For this w we see that $A_\gamma \succeq 0$ and rank $A_\gamma = 3n$. Since $t := t_{\min} + w \in (t_{\min}, t_{\max})$, we know that the other two rank conditions in (19) also hold. This concludes the proof of $(iii) \Rightarrow (ii)$ when $E > 0$.

If $E = 0$, then $Q(w) > 0$ for $0 < w < t_{\max} - t_{\min}$, so $A_{\gamma(t_{\min}+w)} \succ 0$ and β admits a $(3n+1)$–atomic representing measure. At the endpoints, rank $A_{\gamma(t)} = 3n$, but one of A_1, A_2 has rank $3n - 1$. Thus, by [26, Theorem 3.1] and Theorem 2.2, the minimum number of atoms is $3n + 1$ when $E = 0$. For $E > 0$, it is $3n$, since rank $M(n) = 3n$.

Finally, $(ii) \Rightarrow (i)$ is immediate. □

Proof of Theorem 3.2. Since rank $M(n) < 3n$, there must exist an additional column relation that is not generated by the relation $XY = X^3 + q_2 X^2 + q_1 X + q_0$. using recursiveness and linearity. Any such additional relation can be written in the form

$$\sum_{\substack{i,j \in \mathbb{Z}_+, \\ i+j \le n}} \alpha_{i,j} X^i Y^j = \mathbf{0}, \qquad \alpha_{i,j} \in \mathbb{R}. \tag{20}$$

We distinguish two cases according to whether the additional relation involves the monomial Y^n or not.

Case 1: *There exists an additional relation* (20) *with* $\alpha_{0,n} = 0$.

Recall that a column $X^i Y^j$ of $M(n)$ corresponds (via (6) with $m = 3$) to a linear combination $\sum_{t=-j}^{2j} q_{j,t}\, T^{t+i}$ of the columns T^ℓ of the Hankel matrix $A_{\gamma(t)}$. If $(i,j) \neq (0,n)$, then necessarily $j \le n-1$, and hence the exponents $t+i$ range only over $-n+1 \le t+i \le 2n-1$. Therefore, an additional relation (20) with $\alpha_{0,n} = 0$ induces a nontrivial linear dependence among the columns indexed by $\vec{T}^{(-n+1,2n-1)}$, i.e., among the columns of $A_{\widetilde{\gamma}}$. By [26, Theorem 3.1], the sequence $\gamma(t)$ admits an $\mathbb{R}$–representing measure vanishing on $\{0\}$ for some $t \in \mathbb{R}$ if and only if rank $A_{\gamma(t)} = $ rank $A_{\widetilde{\gamma}}$. In particular, this forces

$$\text{rank}\left((A_{\gamma(t)})|_{\vec{T}^{(-n+1,2n)}}\right) = \text{rank } A_{\widetilde{\gamma}}.$$

Arguing as in the proof of Claim 2 in Theorem 3.1, it follows that necessarily

$$t = t_0 = c^T A_{\widetilde{\gamma}}^{\dagger} c \quad \text{and} \quad c \in \text{Ran}(A_{\widetilde{\gamma}}),$$

i.e., $c = A_{\widetilde{\gamma}} w$ for some $w \in \mathbb{R}^{3n-1}$. Finally, Theorem 2.2 together with [26, Theorem 3.1] yields the equivalences stated in Theorem 3.2 in this case.

Case 2: *For every additional relation* (20)*, one has* $\alpha_{0,n} \neq 0$.

In this case, each additional relation involves Y^n nontrivially. Combining such a relation with the relation corresponding to the rational expression

$$y^n = \frac{1}{x^n}\Big(x^3 + q_2x^2 + q_1x + q_0\Big)^n$$

(i.e., the relation coming from $p(X,Y) = \mathbf{0}$) yields a nontrivial linear dependence among the columns T^ℓ of any positive semidefinite completion $A_{\gamma(t)}$ with $\ell \in \{-n, -n+1, \ldots, 2n\}$. Here $A_{\widetilde{\gamma}} \succ 0$ (otherwise Case 1 applies) and rank $M(n) = 3n-1$; the additional relation gives, by Schur complementation, $\Delta := t_{\max} - t_{\min} = -2q_0^n\big(\gamma_n - b^T A_{\widetilde{\gamma}}^{-1} c\big)$. If $\Delta > 0$, the Schur-complement calculation in the proof of Theorem 3.1 gives, at $t = (t_{\min} + t_{\max})/2$, $A_{\gamma(t)} \succeq 0$, rank $A_{\gamma(t)}$ = rank A_1 = rank $A_2 = 3n$, so [26, Theorem 3.1] and Theorem 2.2 give a $3n$–atomic representing measure, corresponding to the second alternative in (iii). This positive semidefinite completion is unique. Every finitely atomic representing measure for β induces this completion, so $3n$ atoms are necessary. If $\Delta < 0$, no positive semidefinite completion exists. It remains to consider $\Delta = 0$, when any positive semidefinite completion has $t = t_{\min} = t_{\max}$. At this value, the Schur complement of $A_{\widetilde{\gamma}}$ is zero, so $A_{\gamma(t)} \succeq 0$ and rank $A_{\gamma(t)}$ = rank $A_{\widetilde{\gamma}}$. However, by [26, Theorem 3.1], in order for $\gamma(t)$ to admit an $\mathbb{R}$–representing measure vanishing on $\{0\}$, there must also be a nontrivial dependence among the columns indexed by $\ell \in \{-n+1, \ldots, 2n\}$ in which T^{2n} appears with a nonzero coefficient. Consequently,

$$\operatorname{rank}\big((A_{\gamma(t)})|_{\vec{T}^{(-n+1,2n)}}\big) = \operatorname{rank}\, A_{\widetilde{\gamma}},$$

and the same Schur-complement argument as in Case 1 yields

$$c = A_{\widetilde{\gamma}} w \text{ for some } w \in \mathbb{R}^{3n-1}, \qquad t_0 = c^T A_{\widetilde{\gamma}}^{\dagger} c, \qquad \operatorname{rank}\, A_{\gamma(t_0)} = \operatorname{rank}\, A_{\widetilde{\gamma}}.$$

Therefore, the equivalences in Theorem 3.2 also follow in this case. □

The following example illustrates our procedure for solving the $\mathcal{Z}(p)$–TMP in the case $m = 3$, with $n = 4$.

Example 3.3. Consider the truncated moment sequence $\beta \equiv \beta^{(8)}$ generated by the 14–atomic measure

$$\mu = \sum_{\ell=1}^{14} \rho_\ell\, \delta_{(x_\ell, y_\ell)},$$

where

$$\rho_\ell = \frac{1}{14}, \quad x_\ell = \ell, \quad y_\ell = \frac{(x_\ell+1)(x_\ell+2)(x_\ell+4)}{x_\ell}, \quad \ell = 1, \ldots, 14.$$

The corresponding moments satisfy, for instance,

$$\beta_{00} = 1, \qquad \beta_{10} = \frac{15}{2}, \qquad \beta_{01} = \frac{88829303}{630630}, \qquad \ldots,$$

$$\beta_{80} = \frac{443370241}{2}, \qquad \ldots,$$

$$\beta_{08} = \frac{22487477336665209271315822126590856880864213410143767741 77}{23730165424120344378453143258050546875 0}.$$

Using *Mathematica* [23], we compute the row-reduced form of the moment matrix $M(4)(\beta)$, which shows that $M(4)(\beta) \succeq 0$ and that β is p–pure for

$$p(x,y) = xy - x^3 - 7x^2 - 14x - 8.$$

Moreover, the columns X^3, X^4, and X^3Y in the column space of $M(4)$ are linearly dependent, and hence rank $M(4) = 12$. Following the procedure from Section 2, we construct the associated univariate strong moment sequence $\gamma \equiv \gamma^{(-8,16)}$ via (9). In particular,

$$\gamma_{-8} = \frac{1400837170807195875714994726099439569487325591594631638817 - 2373016542412034437845314325805054687 50\,\beta_{2,7}}{39812611103619862763169413031925776384000000000},$$

$$\gamma_{-7} = \frac{795732381288691429080031515331406184509}{11048010629265141181920694037053440000000},$$

$$\gamma_{-6} = \frac{44557083929921976202039121208 1493}{61316520308941842501502353408000 00},$$

$$\vdots$$

$$\gamma_{14} = \frac{2405869901763265}{2},$$

$$\gamma_{15} = \frac{32512083310326375}{2},$$

$$\gamma_{16} = \frac{2596336578534357052143750\,\beta_{2,7} - 14754296464684589107824850551429749877576317}{259633657853435705214375 0},$$

where $\beta_{2,7}$ plays the role of a free parameter appeared only in γ_{-8} and γ_{16}.

A direct computation shows that there is no choice of $\beta_{2,7}$ for which

$$\text{rank } A_{(\gamma_{-8},\ldots,\gamma_{16})} = \text{rank } A_{(\gamma_{-8},\ldots,\gamma_{14})} = \text{rank } A_{(\gamma_{-6},\ldots,\gamma_{16})}.$$

Thus, the remaining possibility is that $A_\gamma \succ 0$, which leads to the admissible interval

$$5.9031917636064208814 \times 10^{18} < \beta_{2,7} < 5.9031917636066715225 \times 10^{18}. \quad (21)$$

In this regime, β admits infinitely many 13–atomic representing measures. At the endpoints of (21), one has $A_\gamma \succeq 0$, rank $A_\gamma = 12$, and A_γ is recursively generated in both directions. For instance, for

$$\beta_{2,7} \approx 5.9031917636064208814 \times 10^{18},$$

the zeros of the generating function are approximately

$$\begin{aligned}
&\widetilde{t}_1 \approx 9.35449 \times 10^{-11}, && \widetilde{t}_2 \approx 1, && \widetilde{t}_3 \approx 2.00001,\\
&\widetilde{t}_4 \approx 3.00159, && \widetilde{t}_5 \approx 4.03688, && \widetilde{t}_6 \approx 5.23594,\\
&\widetilde{t}_7 \approx 6.70231, && \widetilde{t}_8 \approx 8.37317, && \widetilde{t}_9 \approx 10.0875,\\
&\widetilde{t}_{10} \approx 11.6566, && \widetilde{t}_{11} \approx 12.9415, && \widetilde{t}_{12} \approx 13.9981.
\end{aligned}$$

Consequently, β admits a 12–atomic representing measure

$$\sum_{\ell=1}^{12} \widetilde{\rho}_\ell\, \delta_{(\widetilde{t}_\ell, \widetilde{s}_\ell)}, \qquad \widetilde{s}_\ell = \frac{(\widetilde{t}_\ell + 1)(\widetilde{t}_\ell + 2)(\widetilde{t}_\ell + 4)}{\widetilde{t}_\ell},$$

which differs from the original generating measure μ.

4. More concrete solutions to the TMP on $p(x, y) = xy - x^4 - q_3x^3 - q_2x^2 - q_1x - q_0,\ q_0 \neq 0$

In this section, we derive more explicit numerical conditions for solving the $\mathcal{Z}(p)$–TMP, where p is as specified in the section title. Starting from the equivalence established in Theorem 2.2, we express the $\mathcal{Z}(p)$–TMP as a Hankel positive semi-definite completion problem with three free moments. The main results are as follows: Theorem 4.1 characterizes the existence of a positive definite completion of the Hankel matrix from Section 2 in terms of a system of inequalities. Theorem 4.2

addresses the complementary case where no positive definite completion exists and resolves the $\mathcal{Z}(p)$–TMP in that setting.

Setup: the induced univariate sequence and its free moments. Assume the notation from Section 2 and let $\gamma \equiv \gamma^{(-2n,6n)}$ be the sequence defined by (9) for $m = 4$. From (4) we obtain

$$h(k) = \max\{2n + 2(k-1), 3k\} + 1 = \begin{cases} 2n + 2k - 1, & \text{if } k \le 2n-2, \\ 6n - 2, & \text{if } k = 2n-1, \\ 6n + 1, & \text{if } k = 2n. \end{cases}$$

Hence,

$$\mathcal{I} = \{-2n, -2n+1, \dots, 6n-4, 6n-2\},$$

and the free moments in γ are γ_{6n-3}, γ_{6n-1}, and γ_{6n}.

We order the moments γ_s as

$$\begin{aligned} &\underbrace{\gamma_0, \gamma_1, \dots, \gamma_{2n}}_{j=0}, \underbrace{\gamma_{-1}, \gamma_{2n+1}, \gamma_{2n+2}}_{j=1}, \dots, \underbrace{\gamma_{-k}, \gamma_{2n+2k-1}, \gamma_{2n+2k}}_{j=k}, \cdots \\ &\qquad \cdots, \underbrace{\gamma_{-2n+2}, \gamma_{6n-5}, \gamma_{6n-4}}_{j=2n-2}, \underbrace{\gamma_{-2n+1}, \gamma_{6n-2}}_{j=2n-1}, \underbrace{\gamma_{-2n}}_{j=2n}. \end{aligned} \tag{22}$$

The auxiliary moments are γ_{-2n+1}, γ_{6n-2}, and γ_{-2n}.

To illustrate the recursion, for $j = 0$ we have

$$\gamma_0 = \beta_{0,0}, \quad \gamma_1 = \beta_{1,0}, \quad \dots, \quad \gamma_{2n} = \beta_{2n,0}.$$

For $j = 1$ one obtains

$$\begin{aligned} \gamma_{-1} &= \frac{1}{q_0}\Big(\beta_{0,1} - \sum_{t=0}^{3} q_{1,t}\gamma_t\Big), \\ \gamma_{2n+1} &= \beta_{2n-2,1} - \sum_{t=-1}^{2} q_{1,t}\gamma_{t+2n-2}, \\ \gamma_{2n+2} &= \beta_{2n-1,1} - \sum_{t=-1}^{2} q_{1,t}\gamma_{t+2n-1}, \end{aligned}$$

and similarly for intermediate $j = k$. For the last two blocks one has

$$\begin{aligned} \gamma_{-2n+1} &= \frac{1}{q_0^{2n-1}}\Big(\beta_{0,2n-1} - \sum_{t=-2n+2}^{6n-4} q_{2n-1,t}\gamma_t - \gamma_{6n-3}\Big), \\ \gamma_{6n-2} &= \beta_{1,2n-1} - \sum_{t=-2n+1}^{6n-5} q_{2n-1,t}\gamma_{t+1} - q_{2n-1,6n-4}\gamma_{6n-3}, \\ \gamma_{-2n} &= \frac{1}{q_0^{2n}}\Big(\beta_{0,2n} - \sum_{t=-2n+1}^{6n-4} q_{2n,t}\gamma_t - q_{2n,6n-3}\gamma_{6n-3} - q_{2n,6n-2}\gamma_{6n-2} - \\ &\qquad q_{2n,6n-1}\gamma_{6n-1} - \gamma_{6n}\Big). \end{aligned}$$

Parametrization of the Hankel matrix. Introduce free parameters t_1, t_2, t_3 by

$$t_1 := \gamma_{6n-3}, \qquad t_2 := \gamma_{6n-1}, \qquad t_3 := \gamma_{6n},$$

and set $\underline{t} := (t_1, t_2, t_3)$. The corresponding Hankel matrix $A_{\gamma(\underline{t})}$ has the form

$$\begin{array}{c} \\ T^{-n} \\ T^{-n+1} \\ \vdots \\ 1 \\ \vdots \\ T^{3n-2} \\ T^{3n-1} \\ T^{3n} \end{array} \begin{array}{c} \begin{array}{ccccccccc} T^{-n} & T^{-n+1} & \cdots & 1 & \cdots & T^{3n-2} & T^{3n-1} & T^{3n} \end{array} \\ \begin{pmatrix} \gamma_{-2n}(\underline{t}) & \gamma_{-2n+1}(t_1) & \cdots & \gamma_{-n} & \cdots & \gamma_{2n-2} & \gamma_{2n-1} & \gamma_{2n} \\ \gamma_{-2n+1}(t_1) & \gamma_{-2n+2} & \cdots & \gamma_{-n+1} & \cdots & \gamma_{2n-1} & \gamma_{2n} & \gamma_{2n+1} \\ \vdots & \vdots & & \vdots & & \vdots & \vdots & \vdots \\ \gamma_{-n} & \gamma_{-n+1} & \cdots & \gamma_0 & \cdots & \gamma_{3n-2} & \gamma_{3n-1} & \gamma_{3n} \\ \vdots & \vdots & & \vdots & & \vdots & \vdots & \vdots \\ \gamma_{2n-2} & \gamma_{2n-1} & \cdots & \gamma_{3n-2} & \cdots & \gamma_{6n-4} & t_1 & \gamma_{6n-2}(t_1) \\ \gamma_{2n-1} & \gamma_{2n} & \cdots & \gamma_{3n-1} & \cdots & t_1 & \gamma_{6n-2}(t_1) & t_2 \\ \gamma_{2n} & \gamma_{2n+1} & \cdots & \gamma_{3n} & \cdots & \gamma_{6n-2}(t_1) & t_2 & t_3 \end{pmatrix} \end{array},$$

where

$$\begin{aligned} \gamma_{-2n}(\underline{t}) &:= C + Dt_1 + Et_2 - q_0^{-2n} t_3, \\ \gamma_{-2n+1}(t_1) &:= F - q_0^{-2n+1} t_1, \\ \gamma_{6n-2}(t_1) &:= G - Ht_1. \end{aligned}$$

For integers $i \le j$ we write

$$\vec{T}^{(i,j)} := \begin{pmatrix} T^i & T^{i+1} & \cdots & T^j \end{pmatrix}.$$

4.1. **Existence of a positive definite completion.** We characterize when there exist t_1, t_2, t_3 such that $A_{\gamma(\underline{t})}$ is positive definite. By [4, Theorem 3.9] together with Theorem 2.2, positive definiteness of $A_{\gamma(\underline{t})}$ is sufficient for β to admit a $\mathcal{Z}(p)$–representing measure.

Assume that

$$(A_{\gamma(\underline{t})})|_{\vec{T}^{(-n+1,3n-2)}} \succ 0. \tag{23}$$

We proceed in three steps.

Step 1: extending to $F_1(t_1)$. Consider

$$F_1(t_1) := (A_{\gamma(\underline{t})})|_{\vec{T}^{(-n+1,3n-1)}}. \tag{24}$$

Then

$$p(t_1) := \det(F_1(t_1)) = c_2 t_1^2 + c_1 t_1 + c_0, \qquad c_0, c_1, c_2 \in \mathbb{R},$$

and (23) implies $c_2 < 0$. A necessary condition for a positive definite completion is that

$$p(t_1) \text{ has a real zero.} \tag{25}$$

Assume that (25) holds, and let α_1, α_2 be the real roots of p, with $\alpha_1 \le \alpha_2$ (where α_1 and α_2 may be equal). Then $F_1(t_1) \succ 0$ for $t_1 \in (\alpha_1, \alpha_2)$ and $F_1(\alpha_i) \succeq 0$ but not definite for $i = 1, 2$. Thus we seek

$$t_1 \in (\alpha_1, \alpha_2) \tag{26}$$

for which $A_{\gamma(\underline{t})} \succ 0$ can be achieved by suitable t_2 and t_3.

Step 2: extending to $F_2(\underline{t})$. Define

$$F_2(\underline{t}) := (A_{\gamma(\underline{t})})|_{\vec{T}^{(-n+1,3n)}}. \tag{27}$$

By Theorem 1.2,

$$F_2(\underline{t}) \succeq 0 \iff t_3 \geq \begin{pmatrix} z_1^T & \gamma_{6n-2}(t_1) & t_2 \end{pmatrix} (F_1(t_1))^{-1} \begin{pmatrix} z_1 \\ \gamma_{6n-2}(t_1) \\ t_2 \end{pmatrix}, \tag{28}$$

where

$$z_1 := \begin{pmatrix} \gamma_{2n+1} & \gamma_{2n+2} & \cdots & \gamma_{6n-4} & t_1 \end{pmatrix}^T.$$

To compute $(F_1(t_1))^{-1}$, set

$$\begin{aligned} \widetilde{\gamma} &:= (\gamma_{-2n+2}, \gamma_{-2n+3}, \ldots, \gamma_{6n-4}), \\ c_1 &:= \begin{pmatrix} \gamma_{2n} & \gamma_{2n+1} & \cdots & \gamma_{6n-4} & t_1 \end{pmatrix}^T, \\ w_1 &:= \gamma_{6n-2}(t_1) - c_1^T A_{\widetilde{\gamma}}^{-1} c_1. \end{aligned}$$

Then

$$\big(F_1(t_1)\big)^{-1} = \begin{pmatrix} A_{\widetilde{\gamma}}^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{w_1} \begin{pmatrix} A_{\widetilde{\gamma}}^{-1} c_1 c_1^T A_{\widetilde{\gamma}}^{-1} & -A_{\widetilde{\gamma}}^{-1} c_1 \\ -c_1^T A_{\widetilde{\gamma}}^{-1} & 1 \end{pmatrix}.$$

Substituting the above equation into (28), we obtain that $F_2(\underline{t}) \succ 0$ is equivalent to

$$\begin{aligned} t_3 > \begin{pmatrix} z_1^T & \gamma_{6n-2}(t_1) \end{pmatrix} \left(A_{\widetilde{\gamma}}^{-1} + \frac{1}{w_1} A_{\widetilde{\gamma}}^{-1} c_1 c_1^T A_{\widetilde{\gamma}}^{-1} \right) \begin{pmatrix} z_1 \\ \gamma_{6n-2}(t_1) \end{pmatrix} \\ - \frac{2t_2}{w_1} \begin{pmatrix} z_1^T & \gamma_{6n-2}(t_1) \end{pmatrix} A_{\widetilde{\gamma}}^{-1} c_1 + \frac{t_2^2}{w_1}. \end{aligned} \tag{29}$$

Step 3: extending to $A_{\gamma(\underline{t})}$. Assuming (26) and (29), we apply Theorem 1.2 once more:

$$A_{\gamma(\underline{t})} \succeq 0 \iff \gamma_{-2n}(\underline{t}) \geq \begin{pmatrix} z_2^T & \gamma_{2n} \end{pmatrix} (F_2(\underline{t}))^{-1} \begin{pmatrix} z_2 \\ \gamma_{2n} \end{pmatrix}, \tag{30}$$

with

$$z_2 := \begin{pmatrix} \gamma_{-2n+1}(t_1) & \gamma_{-2n+2} & \cdots & \gamma_{2n-1} \end{pmatrix}^T.$$

Let

$$\begin{aligned} B &:= F_1(t_1), \\ c_2 &:= \begin{pmatrix} \gamma_{2n+1} & \cdots & \gamma_{6n-2}(t_1) & t_2 \end{pmatrix}^T, \\ w_2 &:= t_3 - c_2^T B^{-1} c_2. \end{aligned}$$

Then

$$(F_2(\underline{t}))^{-1} = \begin{pmatrix} B^{-1} & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{w_2} \begin{pmatrix} B^{-1} c_2 c_2^T B^{-1} & -B^{-1} c_2 \\ -c_2^T B^{-1} & 1 \end{pmatrix}.$$

Substituting the above equation into (30), we conclude that $A_{\gamma(\underline{t})} \succ 0$ is equivalent to

$$\gamma_{-2n}(\underline{t}) > z_2^T \left(B^{-1} + \frac{1}{w_2} B^{-1} c_2 c_2^T B^{-1} \right) z_2 - \frac{2\gamma_{2n}}{w_2} z_2^T B^{-1} c_2 + \frac{\gamma_{2n}^2}{w_2}. \tag{31}$$

Combining Steps 1–3 with Theorem 2.2, we obtain sufficient conditions for β to admit a $\mathcal{Z}(p)$–representing measure in the p-pure case, summarized in the next theorem.

Theorem 4.1 (Purely pure case)**.** *Let $p(x,y) = xy - \sum_{s=0}^{4} q_s x^s$ with $q_s \in \mathbb{R}$, $q_0 \neq 0$, and $q_4 = 1$. Let $\beta = \{\beta_{i,j}^{(2n)}\}$, $i,j \in \mathbb{Z}_+$, $i+j \leq 2n$, be a p-pure sequence and assume* (23)*. If there exists $(t_1,t_2,t_3) \in \mathbb{R}^3$ such that* (25)*,* (26)*,* (29)*, and* (31) *hold, then β admits a $\mathcal{Z}(p)$–representing measure.*

4.2. **Existence of a positive semidefinite completion $A_{\gamma(\underline{t})}$ with a $\mathbb{R}$–representing measure vanishing on $\{0\}$.** In this subsection, we study the existence of an $\mathbb{R}$–representing measure for $\gamma(\underline{t})$ vanishing on $\{0\}$ in the case where β is not p-pure or where no triple $(t_1,t_2,t_3) \in \mathbb{R}^3$ satisfies the conditions of Theorem 4.1. The main result is Theorem 4.2 below.

We say that a column relation in $A_{\gamma(\underline{t})}$ of the form

$$\sum_{i=i_1}^{i_2} a_i T^i = \mathbf{0}, \tag{32}$$

where $a_i \in \mathbb{R}$, $-n \leq i_1 < i_2 \leq 3n$, and $a_{i_1} \neq 0, a_{i_2} \neq 0$, **propagates through** $A_{\gamma(\underline{t})}$ if

$$\begin{aligned} \sum_{i=i_1}^{i_2} a_i T^{i-j} &= \mathbf{0} \quad \text{for } j = 1,\dots,n+i_1, \\ \sum_{i=i_1}^{i_2} a_i T^{i+j} &= \mathbf{0} \quad \text{for } j = 1,\dots,3n-i_2, \end{aligned} \tag{33}$$

are also column relations of $A_{\gamma(\underline{t})}$.

Theorem 4.2 (Singular case)**.** *Let $p(x,y) = xy - \sum_{s=0}^{4} q_s x^s$ with $q_s \in \mathbb{R}$, $q_0 \neq 0$, and $q_4 = 1$. Let $\beta = \{\beta_{i,j}^{(2n)}\}$ for $i,j \in \mathbb{Z}_+$, $i+j \leq 2n$, be a sequence such that no triple $(t_1,t_2,t_3) \in \mathbb{R}^3$ satisfies the conditions of Theorem 4.1.*

Let $\gamma(\underline{t})$ be the associated sequence with free moments $t_1,t_2,t_3 \in \mathbb{R}$, and let $A_{\gamma(\underline{t})}$ be the corresponding Hankel matrix. Set

$$C := (A_{\gamma(\underline{t})})|_{\vec{T}(-n+1,3n-2)},$$

$$F_1(t_1) := (A_{\gamma(\underline{t})})|_{\vec{T}(-n+1,3n-1)}, \quad F_2(\underline{t}) := (A_{\gamma(\underline{t})})|_{\vec{T}(-n+1,3n)}.$$

Assume that $\det(F_1(t_1))$ *has real zeroes α_1, α_2, where $\alpha_1 \leq \alpha_2$. Then β admits a $\mathcal{Z}(p)$–representing measure if and only if one of the following cases occurs:*

- *(i)* ***Degeneracy in*** *C. C is positive semidefinite and singular, and a relation of the form* (32) *satisfied in C propagates through $A_{\gamma(\underline{t})}$, where $\underline{t} \in \mathbb{R}^3$ is uniquely determined by* (33)*.*
- *(ii)* ***Boundary case*** *$t_1 = \alpha_1$. C is positive definite, a relation of the form* (32) *holds in $F_1(\alpha_1)$ and propagates through $A_{\gamma(\underline{t})}$, where $(t_2,t_3) \in \mathbb{R}^2$ is uniquely determined by* (33)*.*
- *(iii)* ***Boundary case*** *$t_1 = \alpha_2$. C is positive definite, a relation of the form* (32) *holds in $F_1(\alpha_2)$ and propagates through $A_{\gamma(\underline{t})}$, where $(t_2,t_3) \in \mathbb{R}^2$ is uniquely determined by* (33)*.*
- *(iv)* ***Degeneracy at the level of*** *$F_2(\underline{t})$. C is positive definite, $t_1 \in (\alpha_1,\alpha_2)$, t_3 equals the right-hand side of* (29)*, and a relation of the form* (32)*, which holds in $F_2(\underline{t})$, propagates through $A_{\gamma(\underline{t})}$.*

(v) ***Degeneracy in the full matrix.*** *C is positive definite, $t_1 \in (\alpha_1, \alpha_2)$, t_3 satisfies (29), $\gamma_{-2n}(\underline{t})$ equals the right-hand side of (31), and in the relation of the form (32), which holds in $A_{\gamma(\underline{t})}$, we have $i_1 = -n$ and $i_2 = 3n$.*

Proof. By assumption, there is no $(t_1, t_2, t_3) \in \mathbb{R}^3$ such that $A_{\gamma(\underline{t})} \succ 0$. We distinguish cases according to the point at which positive definiteness fails.

The key observation is that whenever a nontrivial column relation of the form (32) occurs in a positive semidefinite submatrix, the existence of an $\mathbb{R}$–representing measure vanishing on $\{0\}$ is equivalent to the consistency of its propagation in both directions, as described in (33); see [26, Theorem 3.1].

Case 1: Degeneracy in C. Assume $C \succeq 0$ but $C \not\succ 0$. Then there exists a relation (32) among the columns of C, with indices $-n+1 \leq i_1 < i_2 \leq 3n-2$. By the extension principle [12, Proposition 2.4], this relation holds in any positive semidefinite completion $A_{\gamma(\underline{t})}$. Thus, the existence of a representing measure is equivalent to the consistency of the propagated relations (33), proving (i).

Case 2: Boundary point $t_1 = \alpha_1$. Assume $C \succ 0$ and $t_1 = \alpha_1$. Then $F_1(\alpha_1)$ is positive semidefinite but singular, hence admits a relation of the form (32) with $i_2 = 3n-1$. As above, consistency of (33) characterizes the existence of a representing measure, yielding (ii).

Case 3: Boundary point $t_1 = \alpha_2$. This case is analogous to Case 2, since $F_1(\alpha_2)$ is also positive semidefinite but singular. This proves (iii).

Case 4: Degeneracy at $F_2(\underline{t})$. Assume $C \succ 0$ and t_3 attains the equality in (29). Then $F_2(\underline{t})$ is positive semidefinite but singular, and hence admits a relation of the form (32) with $i_2 = 3n$. Again, consistency of the propagation conditions (33) characterizes the existence of a representing measure. This proves (iv).

Case 5: Degeneracy in the full matrix. Assume $C \succ 0$, $t_1 \in (\alpha_1, \alpha_2)$, t_3 satisfies (29), and the equality holds in (31). Then $A_{\gamma(\underline{t})}$ is positive semidefinite but singular, so there exists a relation (32) with $i_1 = -n$. For a representing measure vanishing on $\{0\}$ to exist, the relation must extend across the full index range, i.e., $i_2 = 3n$. Otherwise, forward propagation in (33) would contradict the positive definiteness of $F_2(\underline{t})$. This proves (v). □

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